

Molecular interactions and reactions

Students develop their understanding of the physical and chemical properties of materials, including gases, water and aqueous solutions, acids and bases. Students explore the characteristic properties of water that make it essential for physical, chemical and biological processes on Earth, including the properties of aqueous solutions. They investigate and explain the solubility of substances in water, and compare and analyse a range of solutions. They learn how rates of reaction can be measured and altered to meet particular needs, and use models of energy transfer and the structure of matter to explain and predict changes to rates of reaction. Students gain an understanding of how to control the rates of chemical reactions, including through the use of a range of catalysts.

Learning outcomes

By the end of this unit, students:

- understand how models of the shape and structure of molecules and intermolecular forces can be used to explain the properties of substances, including the solubility of substances in water
- understand how kinetic theory can be used to explain the behaviour of gaseous systems, and how collision theory can be used to explain and predict the effect of varying conditions on the rate of reaction
- understand how models and theories have developed based on evidence from a range of sources, and the uses and limitations of chemical knowledge in a range of contexts
- use science inquiry skills to design, conduct, evaluate and communicate investigations into the properties and behaviour of gases, water, aqueous solutions and acids and bases, and into the factors that affect the rate of chemical reactions
- evaluate, with reference to empirical evidence, claims about chemical properties, structures and reactions
- communicate, predict and explain chemical phenomena using qualitative and quantitative representations in appropriate modes and genres.

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LOW-RES

A low-resolution, artistic photograph of a pond at night. Several dark, silhouetted lotus seed pods (receptacles) are visible, some standing upright and others tilted. The water surface is covered with numerous small, bright, out-of-focus lights that create a bokeh effect. The overall color palette is dominated by deep blues and purples, with the warm glow of the lights providing contrast. A faint, repeating watermark pattern is visible across the image. A semi-transparent pink rectangular box is centered horizontally, containing the text 'LOW-RES' in a bold, red, sans-serif font.

LOW-RES

CHAPTER 12

Intermolecular forces

Covalent molecular substances are the only substances that can be solids, liquids or gases at room temperature. They can be hard or soft, flexible or brittle, sticky or oily—almost any consistency. This broad range of properties makes molecular substances very useful materials. The properties of covalent molecular substances can generally be explained by the strong covalent bonds inside the molecules and much weaker bonds between molecules.

During this chapter, you will become familiar with the valence shell electron pair repulsion (VSEPR) theory and use of this theory to predict the shape and polarity of molecules. This is because knowing the shapes of molecules is vital when it comes to predicting and explaining the type and strength of forces acting between the molecules. You will learn how intermolecular forces determine many of the physical properties of covalent molecular substances and use these predictions to infer information about the nature and strength of bonding between molecules.

In addition, you will learn to identify intermolecular forces, including dipole–dipole forces, hydrogen bonding and dispersion forces.

Science understanding

- the valence shell electron pair repulsion (VSEPR) theory and Lewis structure diagrams can be used to explain, predict and draw the shapes of molecules
- observable properties, including vapour pressure, melting point, boiling point and solubility, can be explained by considering the nature and strength of intermolecular forces within a covalent molecular substance
- the polarity of molecules can be explained and predicted using knowledge of molecular shape, understanding of symmetry and comparison of the electronegativity of atoms involved in the bond formation
- the shape and polarity of molecules can be used to explain and predict the nature and strength of intermolecular forces including dispersion forces, dipole–dipole forces and hydrogen bonding

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LOW-RES

12.1 Shapes of molecules

The shapes of molecules are critical in determining many physical properties of covalent molecular substances. In particular, molecular shape affects vapour pressure, melting point, boiling point and solubility. This is because the shape of a molecule determines how it interacts with other molecules.

For very large molecules, such as DNA, proteins and enzymes, shape plays a key role in how they behave chemically and biologically. For example, the twisted double helix of DNA allows the molecule to coil up tightly so that it fits inside the nucleus of a cell.

To understand the shape of large molecules like DNA, scientists use complex techniques, such as X-ray crystallography, or powerful computer simulations. However, the shape of small molecules can be predicted using a relatively simple model known as the **valence shell electron pair repulsion (VSEPR) theory**. In this section, you will see how VSEPR theory can be used to predict the shape of molecules such as the water molecule shown in Figure 12.1.1.



FIGURE 12.1.1 Water (H_2O) molecules have a distinctive shape that is responsible for many of its properties. The valence shell electron pair repulsion theory accurately predicts the shape of water molecules.

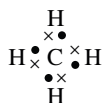


FIGURE 12.1.2 This electron dot diagram (Lewis structure) of a methane (CH_4) molecule shows the four electron pairs surrounding the central carbon.

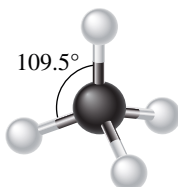


FIGURE 12.1.3 In a methane molecule (CH_4), the electron pairs in the single covalent bonds repel each other. The repulsion forces the bonds as far apart as possible, arranging the hydrogen atoms into a tetrahedral shape at an angle of 109.5° to each other.

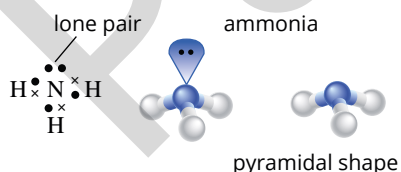


FIGURE 12.1.4 The electron dot diagram of an ammonia (NH_3) molecule shows that the nitrogen atom has one lone pair of electrons and three covalent bonds. These four electron pairs form a tetrahedral arrangement around the atom due to the repulsion of the electron pairs. The result is a pyramidal molecule.

VALENCE SHELL ELECTRON PAIR REPULSION THEORY

As the name suggests, the VSEPR theory uses our knowledge of the valence electrons in the atoms of a molecule to predict the shape of the molecule. The VSEPR theory is based on the principle that negatively charged electron pairs in the outer shell of an atom repel each other. As a consequence, these electron pairs are arranged as far away from each other as possible.

Electron pair repulsion

In general, atoms in covalent molecules are most stable when they have eight electrons in their valence shell (the octet rule). These eight electrons are arranged into four pairs of electrons.

For example, the carbon atom in methane is covalently bonded to four hydrogen atoms. In this arrangement, the carbon atom shares a pair of electrons with each hydrogen atom. The four pairs of electrons give the carbon atom a stable octet as shown in Figure 12.1.2. In this diagram, the dots represent electrons donated to the single covalent bond by the carbon atom. The crosses represent electrons donated to the single covalent bond by the hydrogen atoms. (Note that dots and crosses are used here to distinguish the electrons of the hydrogen atoms from the electrons in the carbon atom valence shell. It is also correct to use dots for all electrons.)

Hydrogen atoms are an exception to the octet rule. Hydrogen atoms are stable with just two electrons because this gives them a full valence shell, or in other words, the stable electron configuration of a helium atom.

The VSEPR theory states that the electron pairs in methane repel each other so that they are as far apart as possible. In effect this means that the angle between the bonds is as large as possible. This is a situation where it might help to imagine the electrons as a 'cloud' of negative charge—remembering that negative charges will always repel other negative charges. This repulsion between the electron pairs results in a tetrahedral shape as shown in the ball-and-stick model in Figure 12.1.3. The tetrahedral shape ensures that the electron pairs are as far from each other as possible with angles of 109.5° between all of the single bonds.

Lone pairs of electrons

Not all electron pairs in molecules exist as covalent bonds. Some electrons form a non-bonding pair of electrons known as a lone pair of electrons. In the VSEPR theory, lone pairs of electrons are treated in the same way as electron pairs in covalent bonds in order to determine the shape of a molecule because they still take up space around the atom.

In the ammonia molecule shown in Figure 12.1.4, the nitrogen atom has a stable octet made up of one lone pair of electrons and three single bonds. The four electron pairs repel each other to form a tetrahedral arrangement. However, the lone pair is

not included when describing the shape of the molecule. Instead, the three hydrogen atoms are described as forming a pyramidal arrangement with the nitrogen atom. The lone pair occupies slightly more space than the bonding electrons (because it is not shared with another atom and is therefore closer to the carbon atom) so the three single covalent bonds are pushed closer together. The bond angle is therefore now slightly less than 109.5° .

In water molecules, the oxygen atom has a stable octet made up of two lone pairs and two single bonds. The four electron pairs repel each other to form a tetrahedral arrangement. This causes the two hydrogen atoms to form a V-shape or bent arrangement with the oxygen atom as shown in Figure 12.1.5. With two lone pairs in the molecule, the two single covalent bonds are pushed closer together. In this case, the bond angle around the central atom is also less than 109.5° .

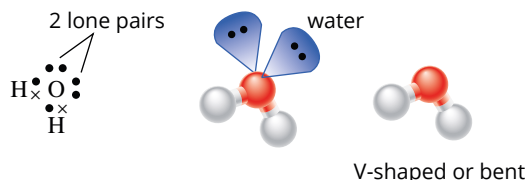


FIGURE 12.1.5 The electron dot diagram of a water (H_2O) molecule shows that the oxygen atom has two lone pairs of electrons and two covalent bonds. These four electron pairs form a tetrahedral arrangement around the atom due to the repulsion of the electron pairs. The result is a V-shaped or bent molecule.

In a hydrogen fluoride molecule, the fluorine atom has a stable octet made up of three lone pairs and one single bond. The four electron pairs repel each other to form a tetrahedral arrangement. Therefore, the hydrogen and fluorine atoms form a linear molecule, as you can see in Figure 12.1.6.

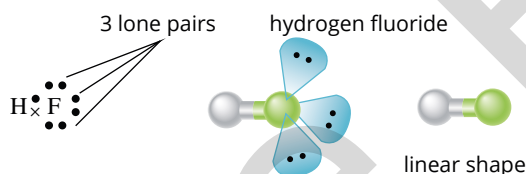


FIGURE 12.1.6 The electron dot diagram of a hydrogen fluoride (HF) molecule shows that the fluorine atom has three lone pairs of electrons and one covalent bond. These four electron pairs form a tetrahedral arrangement around the atom due to the mutual repulsion of the electron pairs. The result is a linear molecule.

Worked example 12.1.1

PREDICTING THE SHAPE OF MOLECULES

Predict the shape of a molecule of phosphine (PH_3).	
Thinking	Working
Draw the electron dot diagram for the molecule.	$\begin{array}{c} \text{H} \times \text{P} \times \text{H} \\ \times \\ \text{H} \end{array}$
Count the number of bonds and lone pairs on the central atom.	There are three bonds and one lone pair.
Determine how the groups of electrons will be arranged to get maximum separation.	Because there are four electron pairs, the groups will be arranged in a tetrahedral arrangement.
Deduce the shape of the molecule by considering the arrangement of just the atoms.	The phosphorus and hydrogen atoms are arranged in a pyramidal shape.

CHEMFILE

That's a-maser-ing!

Ammonia is not just a useful household cleaning product; it can also be used to create microwave lasers—'masers'. A pyramidal ammonia molecule with its hydrogen atoms pointing down can suddenly flip so that the hydrogen atoms are pointing up. This is a bit like an umbrella flipping inside-out on a windy day. When the ammonia molecule flips, it can release microwave radiation.

Charles Hard Townes and James Power Gordon used this property of ammonia to create the first maser, as shown in Figure 12.1.7. Their work laid the foundation for other types of masers, which are now used in radio telescopes to study the far reaches of the universe. Charles Hard Townes was awarded the Nobel Prize in Physics in 1964 for this work.

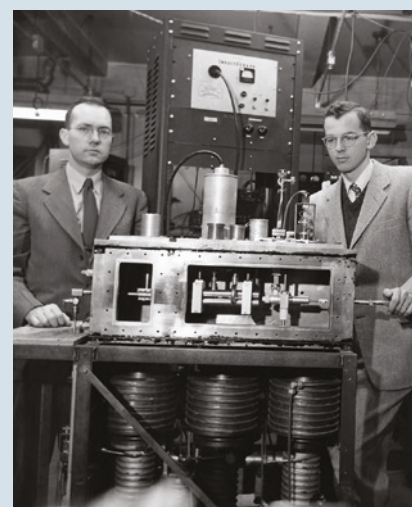


FIGURE 12.1.7 Charles Hard Townes and James Power Gordon invented the first maser using ammonia gas.

i Lone pairs of electrons influence a molecule's shape but are not considered a part of the shape.

Worked example: Try yourself 12.1.1

PREDICTING THE SHAPE OF MOLECULES

Predict the shape of a molecule of hydrogen sulfide (H_2S).

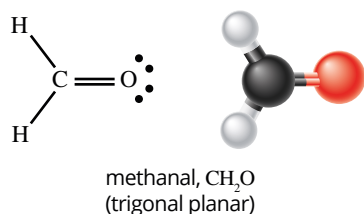


FIGURE 12.1.8 Methanal has a central atom that forms a double bond with an oxygen atom and single bonds with two hydrogen atoms. The bonds repel each other to form a trigonal planar arrangement.



FIGURE 12.1.9 In a carbon dioxide molecule, the carbon atom forms double bonds with two oxygen atoms. The two double bonds repel each other. This results in a linear molecule.

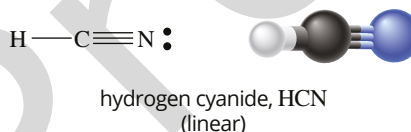


FIGURE 12.1.10 A hydrogen cyanide molecule is linear. In this case, the central carbon atom forms a triple bond with the nitrogen atom and a single bond with the hydrogen atom.

Molecules that do not obey the octet rule

Not all atoms obey the octet rule. For example, boron forms stable molecules with just three electron pairs in its valence shell. In borane (BH_3), the central boron atom is bonded to three hydrogen atoms, but has no lone pairs. There are only three pairs of electrons in this molecule. These three pairs repel each other to form a trigonal planar molecule (Figure 12.1.11).

The VSEPR theory can be used to predict the shape of these molecules. The VSEPR theory predicts that the three electron pairs will move as far away from each other as possible into a trigonal planar shape.

Boron trichloride has a similar arrangement. The three chlorine atoms are located 120° from each other around the central boron atom, as can be seen in Figure 12.1.12.

Beryllium forms stable molecules with just two electron pairs in its valence shell. In beryllium hydride (BeH_2), the central beryllium atom forms single bonds with two hydrogen atoms, but has no lone pairs of electrons. The two single bonds repel each other to form a linear molecule, as shown in Figure 12.1.13.

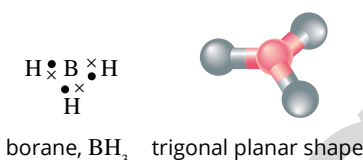


FIGURE 12.1.11 Borane (BH_3) has a central boron atom attached to three hydrogen atoms.

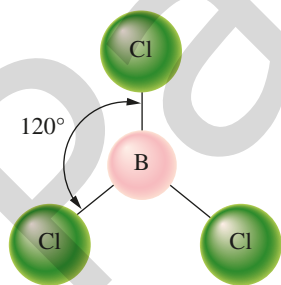


FIGURE 12.1.12 Boron trichloride (BCl_3) has a trigonal planar shape.

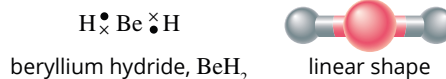


FIGURE 12.1.13 Beryllium hydride has a central beryllium atom attached to two hydrogen atoms. There are only two pairs of electrons in this molecule. These two pairs repel each other to form a linear molecule.

12.1 Review

SUMMARY

- The shapes of molecules play an important role in determining the intermolecular forces between molecules and are related to the physical properties of substances.
- The shapes of simple molecules can be predicted by the valence shell electron pair repulsion (VSEPR) theory as shown in Table 12.1.1.
- The VSEPR theory is based on the principle that electron pairs in the outer shell of an atom repel each other. As a consequence, these electron pairs are arranged as far away from each other as possible.
- Electron pairs can exist in covalent bonds or as lone pairs of electrons. Lone pairs and covalent bonds repel each other.
- Lone pairs influence a molecule's shape but are not considered a part of the shape.
- Double and triple bonds are treated in the same way as single bonds and lone pair electrons in VSEPR theory.

TABLE 12.1.1 Summary of shapes of molecules

Example	Methane, CH ₄	Ammonia, NH ₃	Water, H ₂ O	Hydrogen fluoride, HF
Electron dot diagram				
Number of single bonds	4	3	2	1
Number of lone pairs	0	1	2	3
Ball-and-stick model				
Shape	tetrahedral	pyramidal	V-shaped or bent	linear

KEY QUESTIONS

- Explain VSEPR theory and how it is used to determine the shape of molecules.
- How many electron pairs are there in the valence shell of the fluorine atom in a hydrogen fluoride molecule? State how many single bonds and lone pairs are present.
- Draw the valence structure for each of the following molecules.
 - H₂S
 - HI
 - CCl₄
 - PH₃
 - CS₂
- Describe the shape of each of the molecules in Question 3.
- Copy and complete the table below, determining the shape of each of the molecules with the specific number of bonds listed.

Molecule	Number of single bonds on the central atom	Number of lone pairs of electrons on the central atom	Shape of the molecule
a	4	0	
b	3	1	
c	2	2	

12.2 Properties of covalent molecular substances

The physical properties of covalent molecular substances are primarily determined by the strength of the forces between the molecules. These forces are known as intermolecular forces. The shape of a molecule is one factor that determines the strength of intermolecular forces. However, the covalent bonds inside the molecules also play an important role.

Both the shape of a molecule and the nature of its covalent bonds must be taken into account when predicting the strength of intermolecular forces. This section examines how these factors determine the strength of intermolecular forces and therefore determine important physical properties such as vapour pressure, melting point, boiling point and solubility.

INTERMOLECULAR FORCES

Intermolecular forces exist between molecules and their effects are seen everywhere. If you have ever struggled to flatten out a sheet of cling wrap, then you've experienced the effects of intermolecular forces. Glue is an example of how intermolecular forces can be used to stick things together. The bubbles shown in Figure 12.2.1 are also held together by intermolecular forces.

If you take a closer look at liquids, you will notice further evidence of intermolecular bonding. Intermolecular bonds create the **surface tension** that allows insects to walk on water, as shown in Figure 12.2.2.

Strength of intermolecular forces

Intermolecular forces are 10–100 times weaker than the strong intramolecular forces such as ionic, metallic and covalent bonds. For example, diamond is made up of a lattice of carbon atoms that are bonded with strong covalent bonds. This extremely strong lattice is the reason why diamond has a very high melting point and is the hardest natural substance on Earth.

The atoms in covalent molecules such as water (H_2O) and carbon dioxide (CO_2) are also held together by strong covalent bonds. However, covalent molecular substances tend to have much lower melting and boiling points than ionic, metallic and covalent network substances. This is because the forces *between* the molecules are much weaker. It is the weak intermolecular forces that are broken when a covalent molecular substance is converted from a solid to a liquid or a liquid to a gas.

The vapour pressure of a substance is also related to its intermolecular forces. When a liquid is put into a closed container, some of the particles escape from the surface liquid. These particles form a gas or a vapour that exerts a pressure on the container due to their collisions with the container wall. This pressure is known as the **vapour pressure** (Figure 12.2.3).

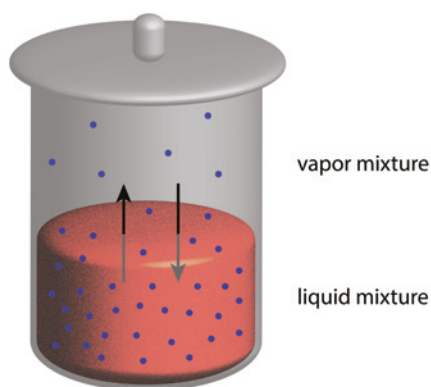


FIGURE 12.2.3 Vapour pressure is the pressure that the gaseous molecules exert on the closed container walls when the rates of evaporation and condensation become equal.

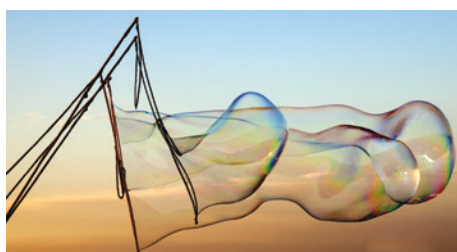


FIGURE 12.2.1 These bubbles are held together by intermolecular forces.

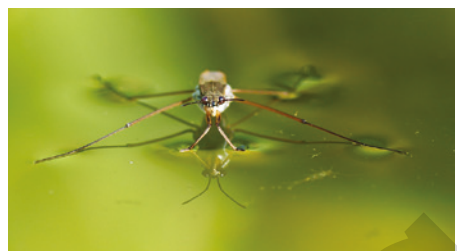


FIGURE 12.2.2 The surface tension created by intermolecular forces allows insects to walk on water.

Liquids with stronger intermolecular forces have lower vapour pressures. For example, water has a vapour pressure at room temperature of 3.17 kPa. On the other hand, the liquid used as a lighter fluid (butane) has a much higher vapour pressure, of approximately 200 kPa. This reflects the fact that the intermolecular forces in water are much stronger than the intermolecular forces in butane. (kPa stands for kilopascals, one of many units used for pressure—normal atmospheric pressure is 101 kPa.)

In a liquid with strong intermolecular forces, the molecules are held together tightly, making it harder for them to escape from the surface of the liquid. Therefore, vapour pressure is also lower.

When a liquid is heated it shows a greater tendency to evaporate as the molecules have a higher average kinetic energy that enables them to overcome the intermolecular forces holding the molecules together. As the temperature increases a greater proportion of the particles will have sufficient kinetic energy to escape from the liquid to produce the gaseous phase and the vapour pressure of a liquid increases (Figure 12.2.4).

The boiling point of a liquid is defined as the temperature at which the liquid's vapour pressure reaches the atmospheric pressure of the surroundings.

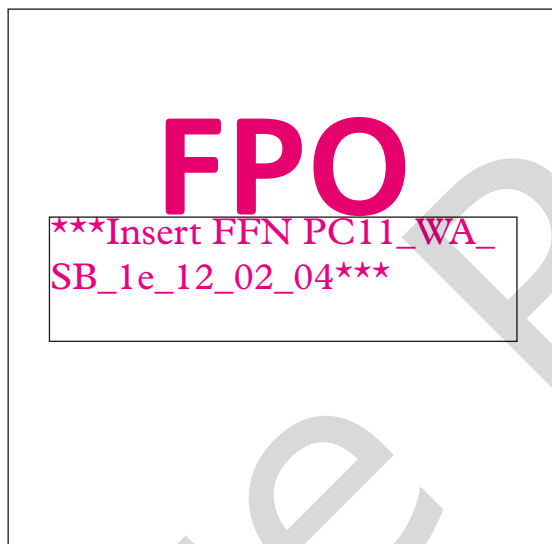


FIGURE 12.2.4 As temperature increases the vapour pressure increases. The boiling points of liquids can be predicted by identifying the temperature at which the vapour pressure is equal to atmospheric pressure. Here, at an atmospheric pressure of 101.3 kPa, ethanol has a boiling point of about 80°C and water about 100°C.

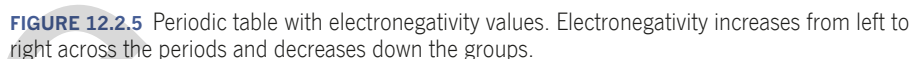
i When liquids change to gases, the intermolecular bonds are broken. For example, when liquid water changes to gaseous water, the intermolecular bonds holding the water molecules to other water molecules are broken, not the covalent bonds between the hydrogen and oxygen atoms.

ELECTRONEGATIVITY AND POLARITY

Intermolecular forces are an example of electrostatic forces. You have already seen many other examples of how electrostatic forces form bonds at the atomic level. The attractions between positive and negative ions in an ionic crystal lattice (Chapter 5) and the attraction of shared electrons and nuclei in a covalent bond (Chapter 6) are all examples of electrostatic forces.

The electrostatic attraction between molecules works in a similar way. In intermolecular forces, the electrostatic attraction is between positive and negative charges in the molecules. These charges appear as a result of uneven electron distributions within the molecules. The following sections examine how the shape of the molecule and the electronegativity of its atoms can cause these uneven electron distributions.

Electronegativity is the key factor that determines the electron distribution in molecules. Electronegativity is the tendency of an atom in a covalent bond to attract electrons. Electronegativity increases from left to right across the periods of the periodic table and decreases down the groups of the table, as shown in Figure 12.2.5. You will remember seeing these patterns in Chapter 3.



When two atoms form a covalent bond, you can regard the atoms as competing for the electrons being shared between them. If the two atoms in a covalent bond are the same (i.e. have identical electronegativities), then the electrons are shared equally between the two atoms. This is the case for diatomic molecules such as chlorine (Cl_2), oxygen (O_2), hydrogen (H_2) and nitrogen (N_2).

Figure 12.2.6 shows the electron distribution in the non-polar fluorine molecule (F_2). The molecule has a high **electron density** between the two fluorine atoms, forming the covalent bond. However, the valence electrons are distributed evenly between the two atoms, making the bond non-polar.

If the covalent bond is between atoms of two different elements, then the electrons will stay closer to the most electronegative atom as it has a stronger pull on the electrons in the bond. An example is the hydrogen fluoride (HF) molecule, shown in Figure 12.2.7. Molecules with an imbalanced electron distribution are said to be **polar**.

A fluorine atom (electronegativity = 3.98) is more electronegative than a hydrogen atom (electronegativity = 2.20). Therefore, in a hydrogen fluoride molecule the electrons tend to stay closer to the fluorine atom. This imbalance in the electron distribution means the fluorine atom is negatively charged and the hydrogen atom

F_2 contains a non-polar bond.

FIGURE 12.2.6 Fluorine molecules have a symmetric distribution of electrons and are therefore non-polar.

is positively charged. A molecule with two oppositely charged poles is said to be polar. The fluorine atom is described as having a partial negative charge, which is represented with the Greek letter delta as δ^- . The hydrogen atom is described as having a partial positive charge, δ^+ . The separation of the positive and negative charges is known as an **electric dipole**, or simply as a **dipole** as they have two oppositely charged poles at each end of the molecule.

Hydrogen fluoride is polar because it has a permanent electric dipole created by the different electronegativities of the two atoms. All diatomic molecules that are made up of atoms with different electronegativities are polar to some extent. The level of **polarity** will depend on the difference between the electronegativities of the two atoms. The greater the difference between the electronegativities, the greater the polarity of the molecule.

The partial charges on polar molecules are different from the charges on ions. They can be considered in terms of the slightly negative end of the bond having a larger share of the electrons, but as the atom still has its normal number of protons in the nucleus, overall this end of the bond can be considered as having an excess of electrons. Conversely, the atom at the slightly positive end of the bond has lost some of its share of the electrons which would have balanced the positive charge of the protons. One way to think about this is that this end is slightly 'naked' in terms of electrons—exposing the positive charge beneath.

The partial charges on a polar molecule will always add to give a total charge of zero, since the charges on polar molecules are due to the unequal sharing of electrons between atoms, so the molecules are still neutral overall. There are no electrons lost or gained as happens in the formation of ions. The partial charges on a polar molecule are also a lot smaller than the charges on ions. That is why intermolecular bonds are much weaker than ionic bonds.

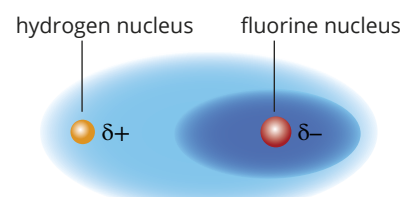
It is not just the covalent bonds in diatomic molecules that can be polar. Covalent bonds in larger molecules can have some degree of polarity. The polarity of any covalent bond can be compared by examining the difference in the electronegativities of the atoms involved in the bond.

Note that if the electronegativity of the atoms bonded together is very similar, the bond can be considered as non-polar, even if there is a small difference in electronegativity. For example, the difference between the electronegativities of carbon and hydrogen is $2.55 - 2.20 = 0.35$. Because this is a relatively small difference, C–H bonds can be considered to be non-polar. This is important as C–H bonds are very common in hydrocarbons and other organic molecules that you will encounter later in the course.

Worked example 12.2.1

COMPARING THE POLARITY OF COVALENT BONDS

Compare the polarity of the covalent bonds in hydrogen fluoride (HF) and carbon monoxide (CO).	
Thinking	Working
Use the table of electronegativity values in Figure 12.2.5 on page xxx to find the electronegativities of the atoms in each molecule.	HF: hydrogen 2.2; fluorine 3.98 CO: carbon 2.55; oxygen 3.44
For each molecule, subtract the lowest electronegativity value from the highest value.	HF: $3.98 - 2.20 = 1.78$ CO: $3.44 - 2.55 = 0.89$
Determine which molecule has the biggest difference in electronegativity in order to determine the more polar bond.	HF is more polar than CO.



HF contains a polar bond; fluorine is more electronegative than hydrogen.

FIGURE 12.2.7 The electron distribution in hydrogen fluoride is asymmetric because of the different electronegativities of the hydrogen and fluorine atoms. Hydrogen fluoride is an example of a polar molecule.

i As the difference in electronegativities of two atoms increases, the covalent bond formed between them increases in polarity. In other words, it becomes more like an ionic bond.

Worked example: Try yourself 12.2.1

COMPARING THE POLARITY OF COVALENT BONDS

Compare the polarity of the covalent bonds in nitrogen monoxide (NO) and hydrogen chloride (HCl).

Polarity of polyatomic molecules

Determining the polarity of polyatomic molecules (those with more than two atoms) is a little more complicated. This is because the polarity of polyatomic molecules depends on the shape of the molecule as well as the polarity of the covalent bonds. As a general rule:

- **symmetrical molecules** are non-polar
- **asymmetrical molecules** are polar.

In this context, it is the symmetry of the polar bonds (dipoles) that is important and symmetry may need to be considered in more than one dimension. One way to consider this is, in the diagrams that follow, imagine that the electrons are being pulled towards the slightly negative end of the dipoles; then consider whether the direction and strength of the individual dipoles (pull of electrons) cancel each other out. If they do, this molecule would be classified as symmetrical and would be non-polar (an analogy for this would be a dead heat in a 'tug-of-war' for electrons). If the dipoles are not balanced out (remembering there might be up to four to consider) then the molecule is considered asymmetrical and the molecule would have a net dipole and be polar.

Non-polar molecules

Even molecules with polar covalent bonds can be non-polar if the molecule is symmetrical.

In methane, the carbon atom is slightly more electronegative than the hydrogen atoms. Therefore, the carbon atom attracts the electrons more strongly than each hydrogen atom and has a partial negative charge, leaving hydrogen with a partial positive charge. However, the methane molecule is a perfect tetrahedron and is therefore symmetrical. The symmetry of the molecule means that the individual dipoles of the covalent bonds cancel each other perfectly (Figure 12.2.8). The result is a molecule with no overall dipole. It is non-polar.

The methane molecule shown in Figure 12.2.8 is an example of a non-polar molecule with polar covalent bonds.

Polar molecules

In asymmetrical molecules, the individual dipoles of the covalent bonds do not cancel each other. This results in a net dipole, making the overall molecule polar.

The chloromethane molecule shown in Figure 12.2.9 is an example of an asymmetric molecule. The chlorine atom is more electronegative than the carbon atom. Therefore, the chlorine atom attracts electrons from the carbon atom while the carbon atom attracts electrons from the hydrogen atoms. The individual dipoles of the covalent bonds are shown in the central image in Figure 12.2.9. These add to give the molecule a net dipole.

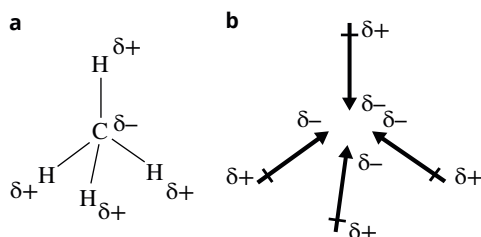


FIGURE 12.2.8 (a) Structure of a methane molecule showing the partial charges on the atoms. (b) The individual dipoles are distributed symmetrically around the molecule.

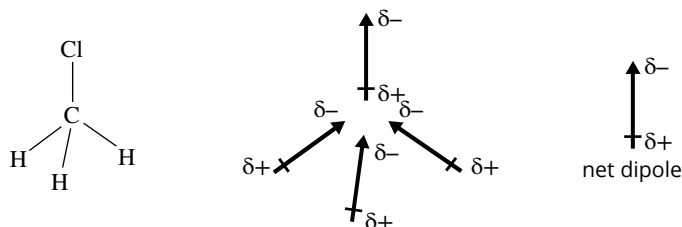


FIGURE 12.2.9 Chloromethane is a polar molecule because the individual dipoles of the covalent bonds add to give a net dipole.

Table 12.2.1 shows some more examples of how symmetry determines the polarity of covalent molecules.

TABLE 12.2.1 Examples of polar and non-polar covalent molecules

Molecule	Valence structure	Symmetrical/asymmetrical	Polar/non-polar
methanal (formaldehyde)	$ \begin{array}{c} \delta+ \\ \text{H} \\ \\ \text{C} = \text{O} \quad \delta- \\ \\ \delta+ \\ \text{H} \end{array} $	asymmetrical	polar
carbon dioxide	$ \begin{array}{c} \delta- \quad \delta+ \quad \delta- \\ \text{O} = \text{C} = \text{O} \end{array} $	symmetrical	non-polar
tetrafluoromethane	$ \begin{array}{c} \text{F} \quad \delta- \\ \\ \delta+ \\ \text{C} \\ / \quad \quad \backslash \\ \delta- \quad \text{F} \quad \delta- \\ \text{F} \quad \delta- \quad \text{F} \end{array} $	symmetrical	non-polar
water	$ \begin{array}{c} \delta- \\ \text{O} \\ / \quad \backslash \\ \text{H} \quad \text{H} \\ \delta+ \quad \delta+ \end{array} $	asymmetrical	polar
Ammonia	$ \begin{array}{c} \delta- \\ \text{N} \\ / \quad \quad \backslash \\ \text{H} \quad \text{H} \quad \text{H} \\ \delta+ \quad \delta+ \quad \delta+ \end{array} $	Asymmetrical	Polar

The polarity of a molecular substance has a significant effect on its solubility. In general, substances that are polar tend to dissolve in polar solvents. Non-polar substances tend to dissolve in non-polar solvents. You can remember this with the saying ‘like dissolves like’. You will learn more about the effect of intermolecular forces on the solubility of substances in Chapter 16.

CHEMFILE

How a microwave oven works

Microwave ovens use the polarity of molecules (especially water molecules) to heat food. The microwave oven irradiates the food with microwaves. The microwaves produce an electric field that interacts with the polar molecules. The electric field causes the molecules to vibrate several million times per second. This gives the molecules extra kinetic energy. As the kinetic energy of the molecules increases, the temperature increases. Therefore, the molecules heat up and cook the food.

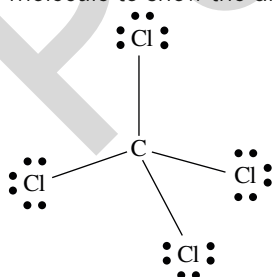
12.2 Review

SUMMARY

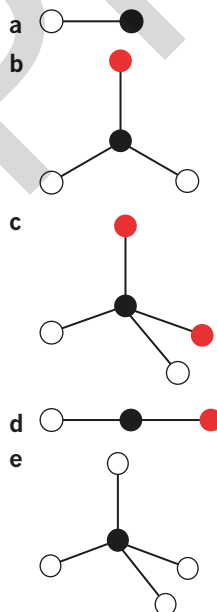
- Many of the physical properties of covalent molecular substances are determined by intermolecular forces. These properties include solubility, melting point, vapour pressure and boiling point.
- Intermolecular forces are broken when liquids change to gas.
- Intermolecular forces are 10–100 times weaker than strong intramolecular forces such as ionic, covalent and metallic bonds.
- Shape and polarity of molecules are factors in determining the strength of intermolecular forces.
- As the difference in electronegativities of two atoms increases, a covalent bond increases in polarity.
- Diatomic molecules containing the same type of atom are non-polar.
- In general, symmetrical molecules are non-polar and asymmetrical molecules with polar bonds are polar.

KEY QUESTIONS

- 1 Covalent bonds can form between the following pairs of elements in a variety of compounds. Use the electronegativity values given in Figure 12.2.5 on page xxx to identify the atom in each pair that would have the largest share of bonding electrons.
 - a S and O
 - b C and H
 - c C and N
 - d N and H
 - e F and O
 - f P and F
- 2 The greater the differences in electronegativity between two atoms, the more polar is the bond formed between them.
 - a Which of the examples in Question 1 would be the most polar bond?
 - b Which of the examples in Question 1 would be the least polar bond?
- 3 Use the electronegativity values in Figure 12.2.5 on page xxx to order the polarity of the covalent bonds between the following atoms from highest to lowest: carbon–nitrogen, nitrogen–nitrogen, hydrogen–nitrogen, sulfur–nitrogen, oxygen–nitrogen.
- 4 Label the following diagram of a tetrachloromethane molecule to show the dipoles between the atoms.



- 5 Examine the symmetry of each of these general diagrams of molecular structures, and determine if the molecules are likely to be polar or non-polar. The white, black and red circles represent atoms with different electronegativities.



- 6 Select the correct description of how the electrons are distributed in a polar, diatomic molecule.
 - A The electrons are located halfway between the two atoms.
 - B On average, the electrons are distributed evenly across the entire molecule.
 - C On average, the electrons spend more time near the most electronegative atom.
 - D On average, the electrons spend more time near the least electronegative atom.

12.3 Types of intermolecular forces

Many factors determine the strength of intermolecular forces, including the size, shape and polarity of molecules. These factors not only determine the strength of the intermolecular forces in a substance, they also determine the types of intermolecular forces.

There are three main types of intermolecular forces:

- dipole–dipole forces
- hydrogen bonding
- dispersion forces.

In this section, you will examine the nature of these three types of intermolecular forces and their role in determining the physical properties of covalent molecular substances.

DIPOLE-DIPOLE FORCES

Dipole–dipole forces only occur in polar molecules. These forces result from the attraction between the positive and negative ends of the polar molecules, as shown in Figure 12.3.1.

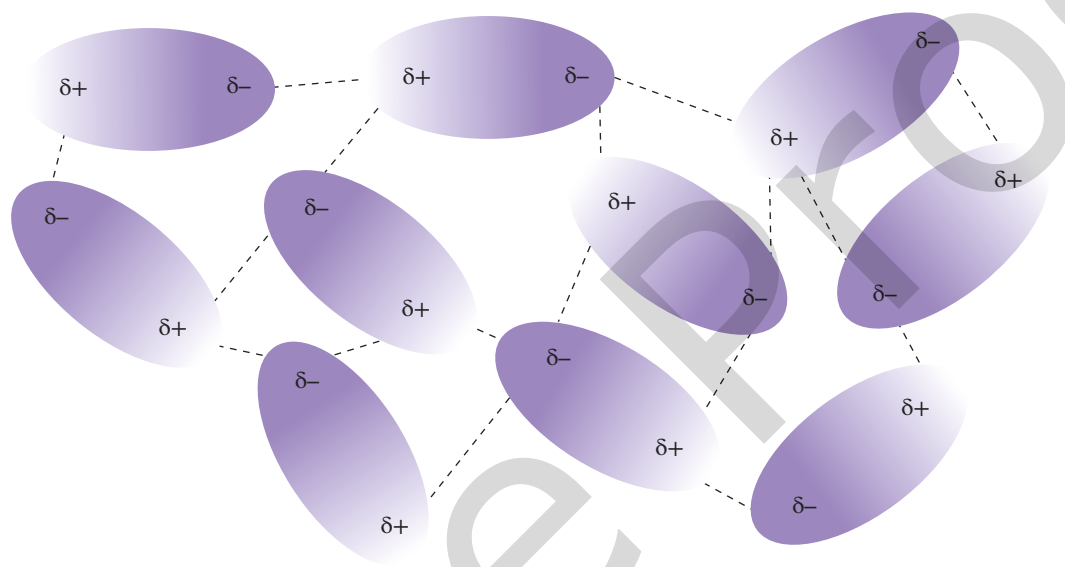


FIGURE 12.3.1 The positive and negative ends of polar molecules attract each other.

Dipole–dipole forces are relatively weak since the partial charges (the $\delta+$ and $\delta-$) on the molecules are small. However, the more polar a molecule is, the stronger the dipole–dipole forces are. The polarity will be larger when there is a large difference in the electronegativities of the atoms or a high degree of asymmetry in the shape of the molecule causing a large imbalance in the bond dipoles.

The strength of the dipole–dipole forces in molecules is directly related to the melting and boiling points of the substance. The stronger the dipole–dipole forces, the higher the melting and boiling points. This is because dipole–dipole forces bond the molecules together in the solid or liquid. Stronger dipole–dipole forces require more energy (i.e. higher temperatures) to break these bonds and change the solid to a liquid or the liquid to a gas.

For example, compare methanal (CH_2O) and ethane (CH_3CH_3), shown in Figure 12.3.2. Methanal is asymmetrical and therefore a polar molecule. This results in dipole–dipole forces between the molecules. Methanal's boiling point is -19°C . On the other hand, the ethane molecule is symmetrical and therefore non-polar. This means there are no dipole–dipole forces and its boiling point is much lower at -88.5°C .

Hydrogen bonding

Hydrogen bonding is a special form of dipole–dipole force. Hydrogen bonding only occurs between molecules in which a hydrogen atom is covalently bonded to an oxygen, a nitrogen or a fluorine atom.

i Molecules are more polar when there is a greater difference in electronegativity of the atoms or a large asymmetry in the shape of the molecule. Molecules that are more polar have stronger dipole–dipole forces between them.

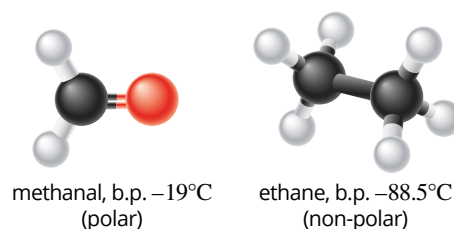


FIGURE 12.3.2 Molecular structures of methanal and ethane

Oxygen, nitrogen and fluorine atoms are small and highly electronegative. Therefore, they strongly attract the electrons in a covalent bond. This creates a significant partial positive charge on a hydrogen atom bonded to these atoms. The partial positively charged hydrogen atom is then attracted to lone pairs of electrons in the nitrogen, oxygen and fluorine atoms of neighbouring molecules, as shown in Figure 12.3.3. Remember that hydrogen only has one electron and this electron is strongly attracted to the N, O or F atom in the covalent bond within the molecule that contains the hydrogen atom. This leaves this partially positive hydrogen atom 'exposed' due to its lack of electrons and it is attracted to a lone pair of electrons on a nitrogen, oxygen or fluorine atom in another molecule.

The result is a relatively strong intermolecular bond that is approximately ten times stronger than a dipole–dipole bond but about one-tenth the strength of an ionic or a covalent bond.

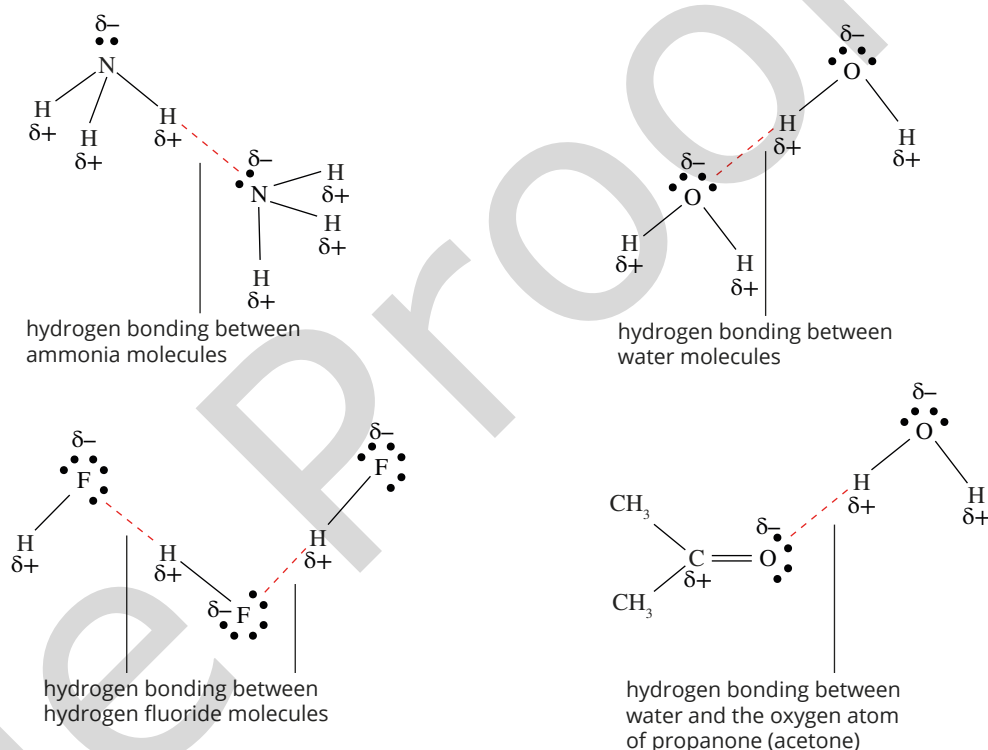


FIGURE 12.3.3 Examples of polar hydrogen bonding between molecules

The presence of hydrogen bonds results in higher melting and boiling points. Figure 12.3.4 demonstrates the effect of hydrogen bonding on boiling point by comparing methanol to ethane and methanal. Recall that ethane is non-polar so it has no dipole–dipole forces between molecules. Methanal has dipole–dipole forces but does not have hydrogen bonding. Methanol has hydrogen bonding between molecules because of the hydrogen atom attached to the oxygen atom. As a result, the boiling point of methanol (64.7°C) is significantly higher than the boiling point of methanal (–19°C) and ethane (–88.5°C).

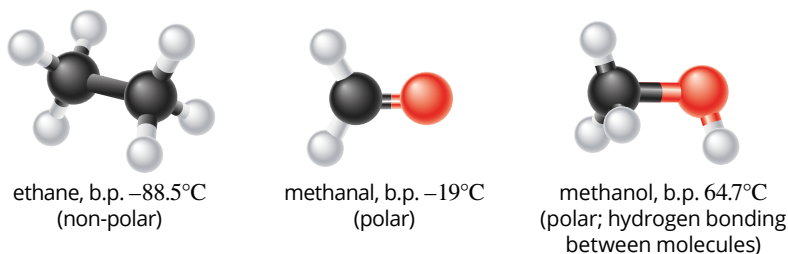


FIGURE 12.3.4 Molecular structures of ethane, methanal and methanol

The two key requirements for hydrogen bonding are a:

- 1 hydrogen atom covalently bonded to an oxygen, a nitrogen or a fluorine atom
- 2 lone pair of electrons on the nitrogen, oxygen or fluorine atoms of neighbouring molecules.

It is both the very high electronegativities and small atomic radii of oxygen, nitrogen and fluorine atoms, combined with the single electron of the hydrogen atom that cause the formation of the hydrogen bonds. Chlorine atoms also have a very high electronegativity but do not form hydrogen bonds because of their larger atomic radius, which reduces the concentration of negative charge around the atom.

Hydrogen bonding in water

Water is a very common substance but it has some unusual properties. One unusual property of water is that ice floats. Ice floats because ice is less dense than liquid water. For most other substances, the solid form is denser than the liquid.

The fact that ice is less dense than liquid water can be explained by hydrogen bonding. Water has two hydrogen atoms attached to an oxygen atom in a V-shape. Therefore, the hydrogen atoms can form hydrogen bonds with the lone pairs of electrons on the oxygen atoms of neighbouring molecules as shown in Figure 12.3.5. In this way, a water molecule can form hydrogen bonds with four other water molecules.

The hydrogen bonding holds the water molecules in ice in a regular crystal lattice. In this lattice, the molecules are held further apart than in liquid water, as shown in Figure 12.3.6. As a result, ice is less dense and therefore floats in liquid water.

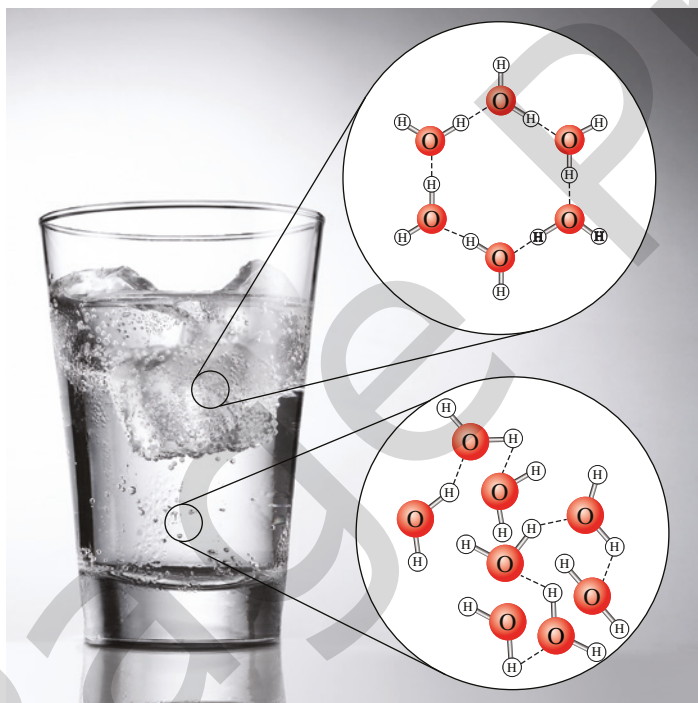


FIGURE 12.3.6 Ice floats because the water molecules form a crystal lattice in which the molecules are spaced more widely apart than in liquid water. This arrangement means ice is less dense than liquid water.

When ice melts and forms liquid water, the density of the water increases rapidly. This is because the open crystal lattice collapses and the water molecules pack together more tightly.

However, as the temperature of the water increases further, water molecules in the liquid move and vibrate more rapidly. The movement causes the molecules to spread further apart. As the molecules move further apart, the liquid becomes less dense.

i Hydrogen bonding occurs in molecules where a hydrogen atom is bonded to an oxygen, a nitrogen or a fluorine atom.

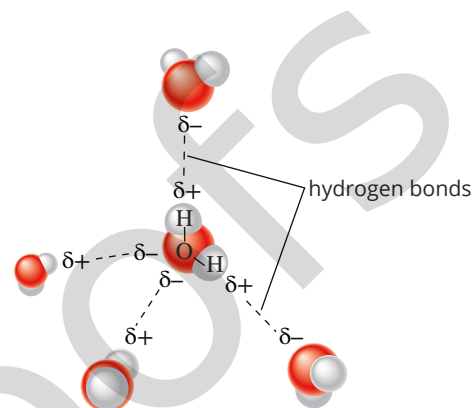


FIGURE 12.3.5 Each water molecule can form hydrogen bonds with four other water molecules. The two lone pairs of electrons on the molecule bond to hydrogen atoms on two neighbouring molecules. In addition, the two hydrogen atoms on the molecule form bonds with lone pairs of electrons in two neighbouring molecules.

CHEMFILE

Snowflake symmetry

No two snowflakes are exactly alike but all snowflakes have something in common. Every snowflake has six symmetrical arms. This phenomenon is the result of hydrogen bonding. The hydrogen bonding between the water molecules creates a crystal lattice with hexagonal symmetry (Figure 12.3.7). As a result, snowflakes show the same hexagonal symmetry as the crystal lattice.



FIGURE 12.3.7 A snowflake with hexagonal symmetry

Dispersion forces

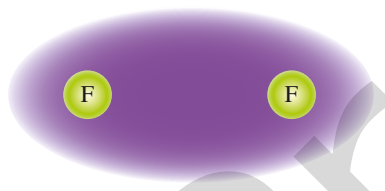
Dipole–dipole forces and hydrogen bonding explain the intermolecular forces in polar substances. However, they do not explain the existence of intermolecular forces in non-polar substances.

We know that intermolecular forces are present in non-polar substances because non-polar substances form liquids and solids. Without intermolecular forces, there would be nothing to hold the molecules together and non-polar substances would only exist as gases. However, there are many non-polar compounds that are liquids at room temperature, such as vegetable oil, and even non-polar solids, such as candle wax. In fact, all non-polar substances form liquids or solids if cooled to a low enough temperature. Even hydrogen liquefies below -259°C .

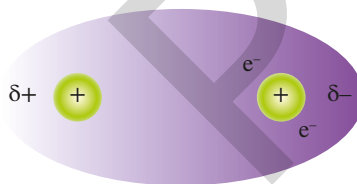
The forces of attraction between non-polar molecules are known as **dispersion forces**. Dispersion forces are caused by **temporary dipoles** in the molecules that are the result of random movement of the electrons surrounding the molecule. These temporary dipoles are also known as **instantaneous dipoles**.

Dispersion forces are always present between molecules, no matter whether they are polar or non-polar as electrons are constantly in motion within atoms (Figure 12.3.8).

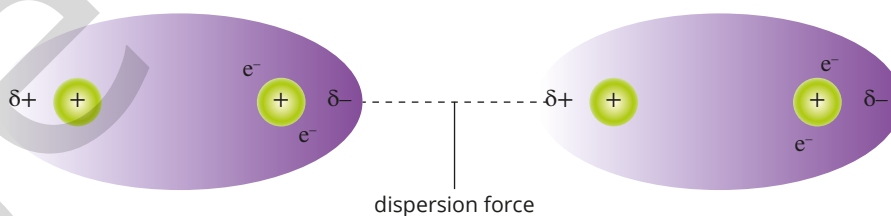
(a) In a molecule, the electrons are constantly moving. In the case of a non-polar molecule, such as fluorine, the electrons spend an equal amount of time around each atom.



(b) Occasionally, the electrons gather more closely together at one end of the molecule, causing one end of the molecule to become negative and the other end to become positive. This is known as a temporary dipole.



(c) These temporary dipoles can then induce (create) dipoles in the neighbouring molecules.



(d) The neighbouring molecules then induce dipoles in their neighbours and so on. The temporary dipoles attract each other to create the intermolecular forces known as dispersion forces.

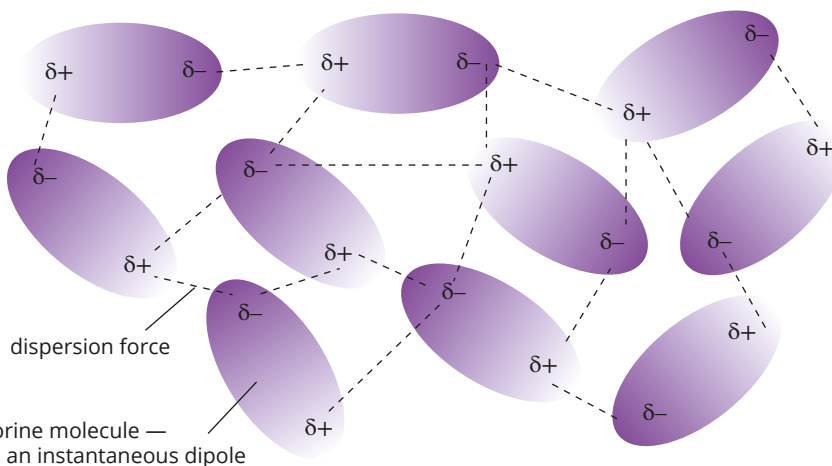


FIGURE 12.3.8 How dispersion forces form within non-polar substances. These diagrams show the forces forming between fluorine molecules.

Strength of dispersion forces

The strength of dispersion forces increases as the size of the molecule increases. Larger molecules have a larger number of electrons. It is easier to produce temporary dipoles in molecules with large numbers of electrons. Since larger molecules have stronger dispersion forces, they have higher melting and boiling points.

Table 12.3.1 shows the boiling points of the halogens (group 17), which all form non-polar, diatomic molecules. The only forces between their molecules are dispersion forces. You can see that as the molecules increase in size and the dispersion forces become stronger, the boiling points of the substances increase.

TABLE 12.3.1 The effect of dispersion forces and the size of atoms on the boiling points of some molecules

Molecule	Molecular mass	Number of electrons	Boiling point (°C)
fluorine (F ₂)	38.0	18	−188
chlorine (Cl ₂)	70.9	34	−35
bromine (Br ₂)	159.8	70	59
iodine (I ₂)	253.8	106	184

The shape of a molecule also influences the strength of the dispersion forces. Molecules that form long chains will tend to have stronger dispersion forces than more compact molecules with similar numbers of electrons.

For example, butane and methylpropane (Figure 12.3.9) both contain four carbon atoms and 10 hydrogen atoms. The boiling point of butane is −0.5°C, while the boiling point of methylpropane is −11°C. The higher boiling point of butane is because of the different shapes of the two molecules; butane is a long molecule while methylpropane is compact. Being less compact and long means butane has more contact area to interact with its neighbouring molecules to form stronger dispersion forces.

It is important to remember that dispersion forces occur between polar molecules as well as non-polar molecules. However, because dispersion forces are generally weaker than dipole-dipole forces, and much weaker than hydrogen bonds, in substances with small polar molecules, dispersion forces do not have a significant influence on the properties of these substances. However, in large molecules, the dispersion forces can dominate over the dipole-dipole forces and hydrogen bonding.

Table 12.3.2 shows the boiling points of three polar molecules: hydrogen chloride (HCl), hydrogen bromide (HBr) and hydrogen iodide (HI). Hydrogen chloride is the most polar of these molecules and therefore has the strongest dipole-dipole forces. However, hydrogen iodide has the highest boiling point because it is a larger molecule and therefore has stronger dispersion forces.

TABLE 12.3.2 Comparison of the boiling points of three polar molecules

Substance	Molecular mass	Number of electrons	Boiling point (°C)
hydrogen chloride (HCl)	36.5	18	−85.1
hydrogen bromide (HBr)	81.0	36	−66.8
hydrogen iodide (HI)	127.9	54	−35.4

The boiling point of hydrogen fluoride (19.5°C) is significantly higher than any of these other compounds. This is because the hydrogen bonding between hydrogen fluoride molecules is much stronger than both the dispersion forces and the dipole-dipole forces between the other molecules listed in Table 12.3.2.

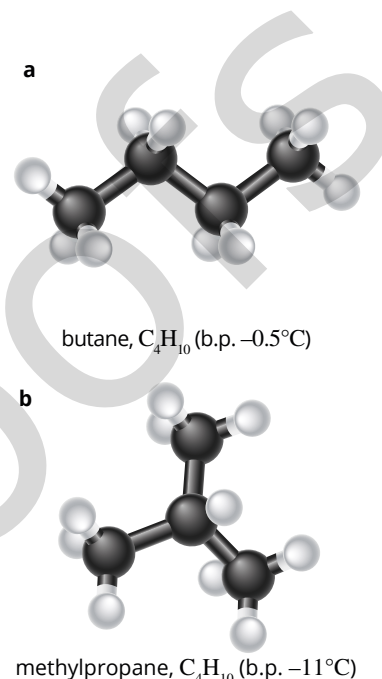


FIGURE 12.3.9 Butane and methylpropane have different boiling points because their molecules are different shapes.

12.3 Review

SUMMARY

- The melting and boiling points of covalent molecular substances increase as the strength of the intermolecular bonding increases.
- There are three main types of intermolecular bonds: dipole–dipole forces, hydrogen bonds and dispersion forces.
- Dipole–dipole forces are only present between polar molecules and are the result of the attraction between the positive and negative ends of polar molecules.
- The polarity of polyatomic molecules depends on the electronegativity of the atoms in the molecule and the symmetry of the molecule.
- The greater the polarity of a molecule, the stronger the dipole–dipole forces.
- Hydrogen bonding occurs between highly polar molecules in which hydrogen atoms are covalently bonded to an oxygen, a nitrogen or a fluorine atom.
- A hydrogen bond is formed between the hydrogen atom in one molecule and lone pairs of electrons in an oxygen, a nitrogen or a fluorine atom in another molecule.
- Hydrogen bonding is the reason why ice is less dense than liquid water.
- Hydrogen bonds are the strongest of the three main types of intermolecular forces.
- Dispersion forces are the result of attraction between temporary dipoles that form in molecules.
- Temporary dipoles are due to random fluctuations in the distribution of electrons in molecules.
- Dispersion forces occur between polar and non-polar molecules.
- Dispersion forces are stronger between larger molecules because it is easier to create temporary dipoles in molecules with a larger number of electrons.

KEY QUESTIONS

- 1 Identify which of the following substances would contain dipole–dipole forces between the molecules: bromine (Br_2), hydrogen chloride (HCl), methane (CH_4), tetrachloromethane (CCl_4), chloromethane (CH_3Cl).
- 2 Refer to Figure 12.2.5 on page xxx for electronegativity values and select which one of the following substances would have the strongest dipole–dipole forces between the molecules in the liquid state.
A fluorine (F_2)
B carbon monoxide (CO)
C hydrogen chloride (HCl)
D methane (CH_4)
- 3 Consider the following substances. Identify whether the molecules are attracted to one another by dispersion forces, dipole–dipole forces or hydrogen bonds.
a NH_3
b CHCl_3
c CH_3Cl
d OF_2
e HBr
f H_2S
g HF
h H_2O
- 4 In ice, each water molecule is surrounded, at equal distances, by four other water molecules. In each case, there is an attraction between the positive hydrogen atom on one water molecule and a lone pair associated with the oxygen atom of another water molecule. Draw a diagram to show the arrangement of four water molecules around another water molecule.
- 5 ‘Cloudy ammonia’ is often used as a cleaning solution in bathrooms. This solution contains ammonia (NH_3) dissolved in water. Draw a diagram to represent hydrogen bonding between a water molecule and an ammonia molecule.
- 6 Identify the types of intermolecular bonding in the following substances.
a methane (CH_4)
b methanol (CH_3OH)
c chloromethane (CH_3Cl)
d methylamine (CH_3NH_2)
e propane ($\text{CH}_3\text{CH}_2\text{CH}_3$)

Chapter review

KEY TERMS

asymmetrical molecule
dipole
dipole–dipole force
dispersion force
electric dipole
electron density

hydrogen bonding
instantaneous dipole
non-polar
polar
polarity
surface tension

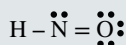
symmetrical molecule
temporary dipole
valence shell electron pair
repulsion (VSEPR) theory
vapour pressure

Shapes of molecules

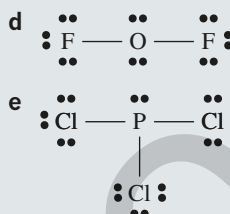
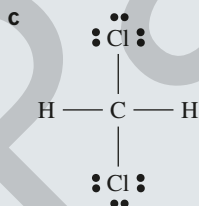
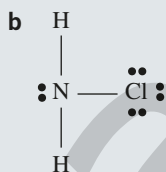
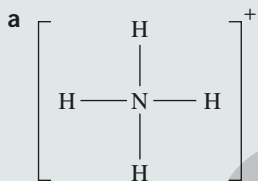
- 1 Match the molecular formula to the correct molecular shape.

Molecular formula	Molecular shape
Phosphorus trichloride (PCl_3)	Tetrahedral
Hypochlorous acid (HOCl)	Linear
Trichloromethane (CHCl_3)	V-shaped
Hydrogen fluoride (HF)	Pyramidal

- 2 Examine the following valence structure and use the VSEPR theory to predict the shape of the molecule.



- 3 How many electrons are involved in the bonding in a molecule of beryllium fluoride?
- 4 All of the following molecules have four pairs of electrons around the central atom. Classify the molecular shapes as tetrahedral, pyramidal or V-shaped.



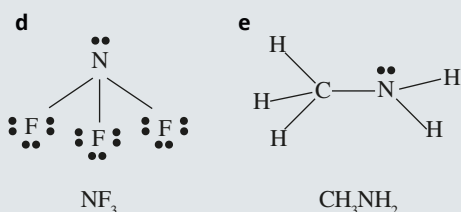
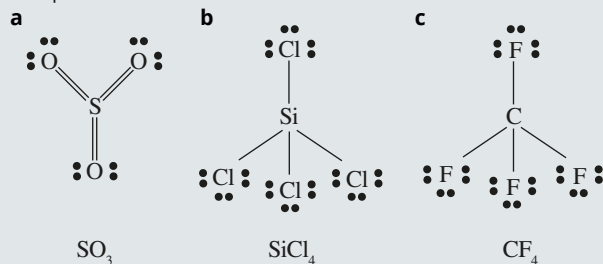
Properties of covalent molecular substances

- 5 Are the following molecules polar or non-polar? Draw structural formulae to help you decide.
- CS_2
 - Cl_2O
 - SiH_4
 - CH_3Cl
 - CH_3CH_3
 - CCl_4
- 6 Use the electronegativities from Figure 12.2.5 on page xxx to determine which of the following molecules contains the most polar bonds.
- CO_2
 - H_2O
 - H_2
 - H_2S
 - NH_3
- 7 For each of the molecules listed in Question 6, state whether they are polar or non-polar.
- 8 Order the following covalent bonds from most polar to least polar.
Si–O, N–O, F–F, H–Br, O–Cl

Types of intermolecular forces

9 For each of the following structures, state whether the:

- molecule is polar or non-polar
- strongest intermolecular forces of attraction between molecules of each type would be dispersion forces, hydrogen bonding or dipole-dipole forces.



10 Consider solid samples of the following compounds. In which cases will the only forces between molecules in the samples be dispersion forces?

- tetrachloromethane (CCl_4)
- sulfur dioxide (SO_2)
- carbon dioxide (CO_2)
- hydrogen sulfide (H_2S)

11 The melting points of four halogens are given in the table below. Describe and explain the trend in melting points of these elements.

Halogen	Melting point ($^{\circ}\text{C}$)
fluorine (F_2)	-220
chlorine (Cl_2)	-101
bromine (Br_2)	-7
iodine (I_2)	114

12 Consider the two compounds OF_2 and CF_4 . OF_2 has a boiling point of -145°C and CF_4 has a boiling point of -128°C . Between molecules of which compound would the intermolecular forces of attraction be greater? Explain your answer.

13 The mass of a hydrogen fluoride molecule is similar to the mass of a neon atom. However, the boiling points of these substances are very different. The boiling point of hydrogen fluoride is 19.5°C whereas that of neon is -246°C . Explain the difference in this property of the two substances.

14 At room temperature, CCl_4 is a liquid whereas CH_4 is a gas.

- Which substance has the stronger intermolecular attractions?
- Explain the difference in the strengths of the intermolecular attractions.

15 Fluorine (F_2) is a gas at room temperature whereas iodine (I_2) is a solid. With respect to intermolecular forces, what factor best explains this difference?

16 Explain the difference between a permanent dipole and a temporary dipole. Your explanation should describe how the dipoles are formed and the type of intermolecular bonding that results.

Connecting the main ideas

17 Consider the following list of molecules:

N_2 , Cl_2 , O_2 , NH_3 , HCl , CH_4 , H_2O , CO_2 , CCl_4 , CHCl_3

a Select one substance from the list above that matches each of the descriptions below. Draw a Lewis diagram for each example, showing bonding and non-bonding electron pairs.

- a molecule that contains one triple bond
- a molecule that contains one double bond
- a molecule that contains two double bonds

b From the list, which substances contain:

- polar molecules?
- symmetrical molecules?
- molecules with hydrogen bonding between them?

18 Water is a polar molecule. Explain how this fact shows that water is not a linear molecule.

19 Explain why covalent molecular substances generally have lower melting and boiling points than ionic and metallic substances.

20 Select the substance that will have the highest boiling point.

- hydrogen fluoride (HF)
- hydrogen chloride (HCl)
- hydrogen bromide (HBr)
- hydrogen iodide (HI)

21 Select the physical property of a substance that is not determined by intermolecular forces.

- boiling point
- vapour pressure
- electrical conductivity
- solubility

22 Select the compound below that has the highest melting point.

- propane (C_3H_8)
- octane (C_8H_{18})
- decane ($\text{C}_{10}\text{H}_{22}$)
- icosane ($\text{C}_{20}\text{H}_{42}$)